

Q1 OR Q2 – Unit 3 (Regression)

Q1

- a) Explain **Bias–Variance trade-off** with diagram. [6]
- b) **SOLVING PROBLEM (VERY HIGH PROBABILITY)** [6]

With following data of company expenditure,
using **linear regression model**, predict expenditure of **6th month**.

x (month) 1 2 3 4 5

y (₹) 12 19 29 37 45

- c) Write short notes on: [6]

- MAE
- RMSE
- R^2

✓ This exact format already appeared in May–Jun 2024 → high repeat chance

Q2

- a) Differentiate between **Lasso and Ridge Regression**. [6]
 - b) Explain **Gradient Descent algorithm** with example. [6]
 - c) Explain **Batch vs Stochastic Gradient Descent**. [6]
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Q3 OR Q4 – Unit 4 (Classification)

Q3

- a) Explain **KNN algorithm** with suitable example. [6]
 - b) Differentiate **Bagging and Boosting**. [6]
 - c) Explain **Random Forest** with diagram. [6]
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Q4

- a) **SOLVING PROBLEM (Metrics – VERY LIKELY)** [6]

Calculate **Precision, Recall and F-score** for given confusion matrix.

- b) Explain **SVM margin and hyperplane**. [6]
 - c) Explain **Multiclass Classification (OvO & OvA)**. [6]
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Q5 OR Q6 – Unit 5 (Unsupervised Learning)

Q5

- a) Explain **K-Means algorithm** with steps and example. [6]
 - b) Explain **Hierarchical clustering** with dendrogram. [6]
 - c) Explain **Elbow method** and **Intrinsic vs Extrinsic metrics**. [6]
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Q6

- a) Explain **DBSCAN algorithm** with advantages. [6]
 - b) Explain **Outlier analysis** with **LOF / Isolation Factor**. [6]
 - c) Compare **K-Means** and **K-Medoids**. [6]
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Q7 OR Q8 – Unit 6 (Neural Networks)

Q7

- a) Explain **ANN architecture**. [6]
 - b) Explain **Backpropagation algorithm** with diagram. [6]
 - c) Explain **Activation functions** (Sigmoid, ReLU). [6]
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Q8

- a) Explain **Multilayer Perceptron** with diagram. [6]
- b) Explain **RNN** with example. [6]
- c) Explain **CNN – Convolution layer**. [6]